

Hidden configuration in a coding agent: a starting-context switch under a fixed model and client version, recovered from per-run telemetry

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Abstract

Evaluations of coding agents treat the model name and the client version as the experimental condition. This paper shows that, for one widely used coding agent, they did not identify it. Over sixteen days in which the resolved model (`claude-sonnet-5`) and the client (Claude Code 2.1.278) were unchanged, the agent’s starting context switched three times between about 19,700 and about 32,000 tokens, alongside changes in its output ceiling. The model name, the version string and the provider’s documentation recorded none of this. The telemetry each run already emits recorded every switch; a daily canary run for the purpose recorded one.

In a daily experiment registered in advance, the agent’s compliance with a project rule moved with the state: the required record was written in 18/18 small-context runs and 11/48 large-context runs. The association is observational. The two states also differed in output ceiling and possibly in serving configuration or model build, the state was never manipulated, and the independent units are eleven days, not 66 runs; within the one week in which the state switched in both directions, the chance of the observed ordering under no association is 1 in 10. A replication registered in advance on the same model reproduced the large-context behaviour but could not test the state. A manipulation registered in advance then set the size directly: on a sibling model in the small-context state, adding about 10,600 tokens of neutral text to the system prompt reproduced none of the large-context pattern, so context size alone does not account for it.

The primary contribution is therefore not that hidden context changes agent behaviour. It is that model and client version labels can fail to identify the experimental condition under which a coding agent runs, and that the failure is detectable from per-run telemetry.

claim	status
The starting context switched three times under a fixed model and client, visible per run	Established from telemetry
Model and version labels do not identify the experimental condition	Established: the same labels covered different measured configurations
A daily outcome registered in advance co-varied with the state	Observational; three small-context days in one series, 1 in 10 within the switching week
Context size alone produces the large-context pattern	Not supported: a registered size manipulation on a sibling model reproduced none of it
Whatever else the switch changed produces the pattern	Not established; the same-model test did not run, and is to be registered again

1. Introduction

Studies of coding agents treat the model name and the client version as the condition. Different studies compare different agents, and a single study compares different instructions or tools, on the assumption that holding those two names fixed holds the condition fixed. This paper reports a case in which it did not, how the case was found, and what it did to the conclusions of the programme that found it.

The programme, registered in advance part by part, asked a practical question: where should a software project place a rule that a coding agent must follow? It compared the same rule placed in the always-loaded project instruction file (`CLAUDE.md`), in an optional skill that the agent may choose to open, and in a seeded folder of past decisions that shows the convention by example. In early October two of its conclusions reversed. In the earlier experiments the skill was opened in nearly every run and the seeded folder was followed in nearly every run. In the later ones the skill was opened in fewer than half the runs, and the seeded folder was followed far less often whenever it contained worked examples. A scoring failure appeared at the same time.

The first explanation was a dated platform change: a daily canary recorded the agent’s cached prefix growing by about 62 per cent on 28 September. The explanation was incomplete in one respect that mattered, because the change was not a single step on one date. Each run’s own telemetry showed that the agent’s starting configuration had switched on 22 September, back on 23 September and again on 28 September, and that the canary had begun after the first two switches.

The primary contribution is not that hidden context changes agent behaviour. It is that model and client version labels can fail to identify the experimental condition, and that this failure is detectable from per-run telemetry. The paper makes three contributions in all, in decreasing order of how firmly they are established. The first, and the main one, is methodological: the same model and client labels covered measurably different configurations, so those labels are insufficient identifiers of an agent’s experimental condition, and three fields of per-run telemetry are enough to recover the difference. The second is an observational co-variation, with its limits stated: a daily outcome registered in advance, and a behavioural signature (how the agent reads files and the format it writes), moved with the configuration, while a negative control, whether the agent completed the task’s code change, did not (section 5.2). The paper does not claim that the starting context causes the behaviour; the states differed in more than context size. The third is a replication on the same model, registered in advance, which reproduced the large-context behaviour but could not test the state, because the state did not switch while it ran. The fourth is a manipulation registered in advance, which set the size of the starting context directly on a sibling model and found that size alone reproduces none of the pattern.

2. Related work

Each statement below about another work was checked on 7 October 2026 against that work’s full text: the arXiv HTML for papers, and the original pages for the postmortem and the issue report. The one direct quotation was matched verbatim.

Behaviour under a fixed model name is known to drift. Chen, Zaharia and Zou compared March and June 2023 snapshots of two chat models and found large changes, including an output-format change that broke their own code-execution check [Chen et al., section 3.5]. That is the same kind of event as the scoring failure in section 5.5; their design has two time points and no visible mechanism. Zhu and Zhang registered measurements on shared endpoints in advance and found them failing their own reliability gates through nondeterminism that no metadata field could identify [Zhu and Zhang]. The present case is the systematic counterpart: a two-state switch that per-run telemetry does identify.

Providers change the layer around the model and disclose little of it. Abraham and Bucknall define system-layer silent updates, covering system prompts, tool access and inference budgets, and note that production

system prompts “are not externally inspectable” [Abraham and Bucknall]. The provider’s April 2026 post-mortem attributes a Claude Code quality regression to harness-side changes that its own evaluations initially missed [Anthropic]. The size of such a change, together with the output ceiling and the tool list, is inspectable per run, and that is enough to date the change and to ask whether behaviour moves with it.

Changes can be detected cheaply from outside. Active black-box probes date API changes at low cost but cannot say which layer changed [Chauvin et al.]. Provider-reported prompt-token counts have been used to expose injected system prompts on open-weight endpoints at a single moment [AgentProv, appendix A.5], and a public issue report used Claude Code’s token telemetry to show server-side context growing between two client versions [claude-code issue 46917]. None of these tracks the signal over calendar time under one fixed client version, observes a switch that reverses, or measures an outcome registered in advance on both sides of it.

The content of an agent’s context drives its behaviour. Wang et al. show that revising a skill’s text changes what agents do, with the skill opened in 88 per cent of Sonnet 4.5 episodes on a fixed open scaffold [Wang et al.]. Mohammadi et al. read per-run token buckets from Claude Code to describe agents’ working sets, without reporting a client version or run dates [Mohammadi et al.]; the present finding suggests that such figures can move with the date.

3. Setting

The agent is Claude Code, run headlessly by a harness that builds a fresh fixture repository for each run, gives a fixed prompt, and archives the run’s full transcript, tool log, skill-invocation counter and provenance (date, requested and resolved model, client version). Throughout the period analysed in section 5 the client was 2.1.278 and the resolved model was `claude-sonnet-5`. The harness requested the alias `sonnet`, which resolved to `claude-sonnet-5` until 6 October and to a different model, `claude-sonnet-5-5`, from that evening (appendix C).

The task is a small web service that needs an API change. The scored obligation is to record the structural decision behind the change in a markdown file carrying a heading and a status. The rule was delivered in one of five ways: not at all (the bare condition); as an installed skill that the agent may open; as a line in `CLAUDE.md`; as a seeded decisions folder with an index and two worked entries; or as the same index with no entries.

Two instruments score whether a record was written: the detector named in the original registrations, and a transcript re-scorer built later to read every route by which the agent writes, including shell commands. Both are reported wherever they differ. An automated check recomputes every count in this paper from the raw run archive (appendix B). The archive, the harness and the registrations are held in a private repository and are not publicly released. A run is excluded only for a recorded reason that does not depend on its outcome, and is marked rather than deleted. On this model and client within the period analysed, 21 runs were excluded: 15 on 3 October, when a transient provider rate limit cut one launch short, and 6 on 28 September, from a launch in which the arms they were to be compared with failed to start. None belongs to the daily series of section 5.2.

The experiments and analyses were conducted largely by an AI agent working under a written research protocol, with the author setting direction and taking the decisions recorded here. Three reviews by separate agent sessions found errors in drafts that the automated checks did not, including both compromised tests in appendix C.

4. The fingerprint

Three fields from each run’s own records suffice. The **starting context** is the input, cache-read and cache-creation tokens of the run’s first model call, which is everything the agent receives before it acts. The **output ceiling** is the maximum output length the run’s result reports. The **tool list** is the set of tools the client announces at start-up.

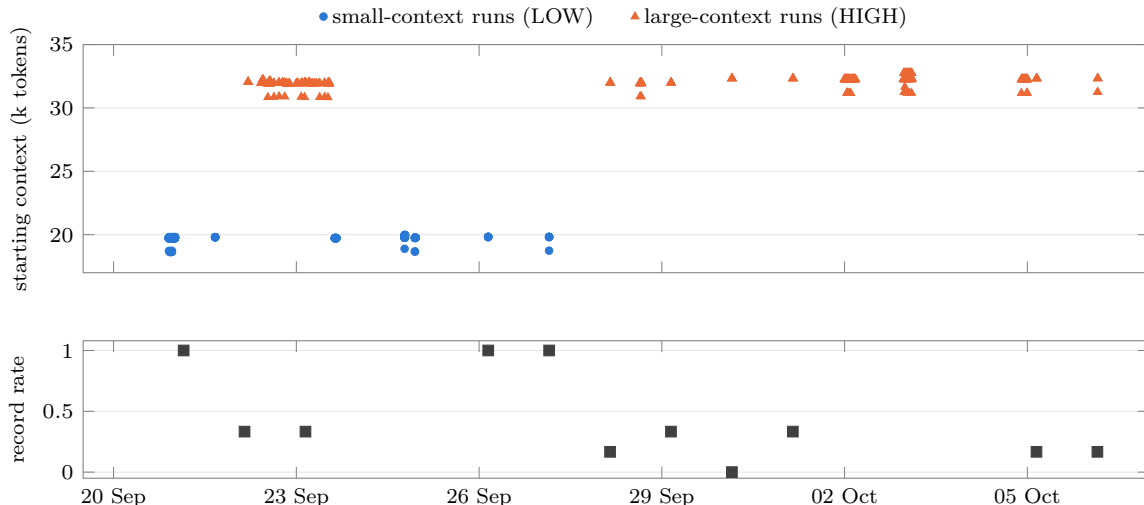


Figure 1: Top: the starting context of every `claude-sonnet-5` run on client 2.1.278 (639 runs), by date. Bottom: in the daily series, the share of each day’s six runs that wrote the decision record (re-scorer). The daily canary began on 23 September at 21:32 UTC, after the first two switches.

A run is classed LOW if its starting context is below 26,000 tokens and HIGH if it is 29,000 or above; runs in between are left unclassified and never read. The boundaries are not a tuning choice: on client 2.1.278, no run in the archive falls between 25,600 and 30,800 tokens.

5. What the archive shows

Everything in this section is post hoc, because the two states were identified on 6 October 2026, after these runs. Each comparison is labelled either a reversal, in which the state switched back and forth within one series, or a before-and-after comparison, in which one switch is confounded with everything else that changed on that date. All tables use runs of `claude-sonnet-5` on client 2.1.278 dated from 15 September 2026.

5.1 The configuration switched

With an identical tool list, the starting context and the output ceiling took three combinations. The LOW state appeared with two different output ceilings: 64,000 tokens on 20 and 21 September, and 128,000 from the evening of 23 September to 27 September. The HIGH state, of about 32,000 tokens with a 64,000 ceiling, appeared on 22 and 23 September and in every run from 28 September to 6 October. Within the sixteen-day archive the starting context therefore switched three times. In a few runs in every state, eight tools from an account-level connector failed to load, which changes the starting context by about 1,100 tokens but not the state. The canary began on 23 September at 21:32 UTC, inside the second LOW period, and recorded a single step on 28 September. The switching continued after the archive window: on the same pinned model, the canary read LOW at 02:50 UTC on 7 October and HIGH at 02:50 on 8 October, and a probe read HIGH at about 15:00 on 7 October.

Figure 1 shows every classified run, and the daily series’ outcome.

5.2 One daily series across three switches

One experiment, registered in advance to measure day-to-day variation, ran the same fixture, prompt and installed skill once a day. The days missing from the series are days on which the launcher’s own pre-flight checks refused to start, not days chosen afterwards. This is the one reversal in the archive.

In LOW the record was written in 18/18 runs and in HIGH in 11/48; the skill was opened in 17/18 and

17/48; and, of the runs that opened it, 17/17 and 11/17 complied. By the originally registered detector, which missed some records written in status formats it could not then read, the records are 14/18 against 8/48.

Within HIGH, the rate before and after 28 September is similar, 4/12 against 7/36, although counts of this size cannot exclude a moderate effect of date. The re-scorer was built after eleven of the twelve days had run, although before the states were identified. Under HIGH both steps are lower: fewer runs open the skill, and fewer of those then follow it.

Statistical treatment. The table below gives each proportion with its exact interval.

state, outcome	proportion	95% CI
LOW, records written	18/18	95% CI [0.815, 1.000]
HIGH, records written	11/48	95% CI [0.120, 0.373]
LOW, skill opened	17/18	95% CI [0.727, 0.999]
HIGH, skill opened	17/48	95% CI [0.222, 0.505]
LOW, complied given opened	17/17	95% CI [0.805, 1.000]
HIGH, complied given opened	11/17	95% CI [0.383, 0.858]

As effect sizes, the LOW minus HIGH difference is 0.77 in records written (Newcombe 95% limits 0.55 and 0.87), 0.59 in skill opened (limits 0.34 and 0.72) and 0.35 in compliance given opening (limits 0.10 and 0.59). These run-level figures treat runs as independent, which they are not, because runs within a day share a configuration and an occasion. The analysis that respects the clustering uses the day as the unit: every LOW day wrote the record in six runs of six, and the eight HIGH days wrote it in 2, 2, 1, 2, 0, 2, 1 and 1 runs of six, a mean day rate of 0.23. The exact one-sided permutation test on day rates gives 1 in 165 over all eleven days with the re-scorer, with no ties; with the registered detector one LOW day ties with three HIGH days and the chance is 4 in 165. Because the LOW days are not spread through the series, a test that respects the switching structure is more informative: within 21 to 27 September, the one week in which the state switched in both directions, the three LOW days outrank the two HIGH days with a chance of 1 in 10. The exact intervals in the table are recomputed from their counts by a script; the Newcombe limits were computed by hand and confirmed by a second, independent computation, and are not checked automatically.

The output-ceiling confound. LOW occurred with two output ceilings, so context size and output ceiling are partly confounded. The first day of the series was LOW with a 64,000 ceiling, the same ceiling as every HIGH day, and wrote the record in 6/6 runs; the two LOW days with the 128,000 ceiling wrote it in 12/12. With the ceiling held at 64,000, the one LOW day exceeds every HIGH day, a chance of 1 in 9 at the day level. This points away from the ceiling as the explanation, but it rests on a single day and does not exclude other serving-side differences that travelled with the state.

Robustness. Four checks, none of which needs new runs. First, a negative control: the agent completed the task’s code change in every run in both states (work happened in 18/18 LOW runs and 48/48 HIGH runs), so the HIGH state did not reduce capability in general; it changed what the agent did beyond the task. Second, leave-one-day-out: dropping any single day leaves the LOW days ranked highest, with a chance of at most 1 in 45 (when a LOW day is dropped) and 1 in 120 (when a HIGH day is dropped), so no single day carries the result. Third, the connector tools: excluding the three runs in which eight connector tools failed to load, every remaining LOW run still wrote the record, and the HIGH rate moves from 0.23 to 0.24. Fourth, the instrument: by the originally registered detector the direction is the same (14/18 against 8/48). These figures and chances are hand-computed from the gated counts.

Effort does not divide the states as cleanly. Mean turns per run were 11.0 in the LOW period with the 64,000 ceiling, 6.2 in the LOW period with the 128,000 ceiling, and 19.8 in HIGH (means computed by hand).

5.3 Across the archive

The same division appears in the skill and seeded-folder arms of other experiments, in a before-and-after comparison.

	LOW	HIGH
skill arm, records	39/39	23/72
skill arm, opened	38/39	29/72
seeded folder with worked entries, records	27/27	11/48

The LOW skill runs include the daily series’ LOW days, which do switch. Every LOW seeded-folder run, however, is dated 20 to 24 September and every HIGH one 2 to 5 October, so that comparison is confounded with the date, the harness version and the mix of launches.

5.4 A behavioural signature

Across every bare run on this model and client from 15 September, which spans 25 experiments and 15 tasks, the agent read files only through shell commands, with no call to the Read tool, in 51/51 LOW runs and in 9/180 HIGH runs. Across every arm with no worked example to copy (index only, table only, directory only and the `CLAUDE.md` rule), the records written in HIGH carried their status as a heading in 80/80 cases, and the records written in LOW carried it as a plain field in 6/6 cases. Both are before-and-after comparisons.

A counterexample bounds the signature. In August, on client 2.1.220, the same model and task ran with a small starting context of about 20,600 tokens, and the agent used the Read tool almost throughout: only 1/21 bare runs and 0/21 skill-arm runs made no Read call. A small context does not by itself produce shell-only reading; the signature belongs to these states under this client.

5.5 Consequences for one programme’s conclusions

Four results of the programme were measured only in HIGH: that the seeded folder was followed far less often than the skill once each had arrived; that worked examples suppressed the seeded folder; a product decision built on those two; and a scoring failure in which the detector read one compliant run in 120, because the agent wrote its status as a heading, the form the detector could not then read. The heading form co-varies with the state (section 5.4), although the detector’s blind spots were not specific to it, since it also missed list-form records on the daily series’ first, LOW, day. Whether these results hold in LOW is untested on this model.

The archive does not show what the extra 12,000 or so tokens contain, since transcripts do not keep the injected context; whether the switch is per account or global; any mechanism; or anything about other model families, since no other model ran in HIGH.

6. Prospective tests

6.1 A same-state replication on the original model

Two earlier attempts to test the state prospectively were compromised, one by the alias change and one by a prediction committed after its data; both are reported in appendix C. The test described here was registered in advance, in a timestamped commit made before its runs (appendix B). It pinned the original model, `claude-sonnet-5`, by full name, and refused to launch if a probe resolved to any other model. It ran five arms of 24 runs each, interleaved in eight blocks, read each run by its own state, and stated the archive’s prediction for both states. Every run resolved to `claude-sonnet-5`, on client 2.1.292, and every run was HIGH. The two instruments agreed on every run.

arm	records written	95% CI
bare	0/24	95% CI [0.000, 0.142]
seeded folder with worked entries	8/24	95% CI [0.156, 0.553]
index, no entries	21/24	95% CI [0.676, 0.973]
CLAUDE.md	24/24	95% CI [0.858, 1.000]
skill: record written	14/24	95% CI [0.366, 0.779]
skill: opened	15/24	95% CI [0.406, 0.812]

The registered reading table required at least 20 scored runs per arm and a bare floor of at most 2 of 24 runs, and, for the HIGH prediction, the seeded folder at most 8 of 24, the skill opened in at most 16 of 24, and CLAUDE.md at least 20 of 24. All were met, the seeded folder exactly at its threshold. The suppression by worked entries replicated as a registered secondary contrast: 21/24 for the index alone against 8/24 with the entries, a two-sided Fisher p of 0.00028, computed by hand.

The large-context findings therefore reproduce on their original model under a newer client, and the compromised test’s departure from them (appendix C) is not evidence against them. The replication does not test the state. An account in which behaviour changed around 28 September for any reason and stayed changed predicts the same result, and only a LOW run on the same model can separate the two. That test was registered to launch automatically when an hourly probe of the pinned model read LOW. From 21:07 UTC on 7 October to 17:59 UTC on 8 October every probe, and the canary of 8 October, read HIGH; the only LOW reading of the pinned model since 6 October is the canary of 7 October. The automatic launch was then stopped, because the test used the seeded folder that is being replaced (section 7), and the test is to be registered again on the replacement.

6.2 A manipulation of context size

The replication could not separate the state from anything else that changed with it, and the archive’s model did not return to LOW. A manipulation therefore set the size of the starting context directly. It was registered in advance on `claude-sonnet-5-5`, which read LOW (21,583 tokens, 128,000 ceiling) while the archive’s model read HIGH. Three arms (bare, the seeded folder with worked entries, the installed skill) ran with and without a fixed block of neutral public-domain prose appended to the system prompt: the opening of Chapter I of Darwin’s *On the Origin of Species*, 33,749 characters, calibrated to add 10,571 tokens and move a LOW starting context into the archive’s HIGH range. Padded and unpadded runs were interleaved in every block, so occasion, client and serving state were shared within each pair. A check of the manipulation confirmed that every padded run started at 30,505 to 32,428 tokens and every unpadded run at 19,934 to 21,857, and every run’s base state was LOW. Every run resolved to `claude-sonnet-5-5` on client 2.1.294, and the two instruments agreed on every run.

arm	records written, unpadded	95% CI
bare	0/24	95% CI [0.000, 0.142]
seeded folder	24/24	95% CI [0.858, 1.000]
skill	15/24	95% CI [0.406, 0.812]

arm	records written, padded	95% CI
bare	0/24	95% CI [0.000, 0.142]
seeded folder	22/24	95% CI [0.730, 0.990]
skill	14/24	95% CI [0.366, 0.779]

The skill was opened in 14/24 unpadded runs and 12/24 padded runs. The registered verdict is **SIZE DOES NOT REPRODUCE IT**: the padded seeded folder stayed at or above 18 of 24, and skill opening moved

by no more than 3 runs in either direction (it moved by 2). Padding moved none of the secondary measures either. The agent read files only through shell commands in every bare run with and without padding (24/24 and 24/24), and mean turns stayed between 4 and 6 in every arm.

The result rules out the simplest mechanism on this model. Adding enough neutral context to reach the HIGH range did not produce the HIGH pattern. If the archive’s model behaves the same way, and if the archive’s switch caused the change in behaviour at all, it did so through something other than size: the content of the added context, the serving configuration, or the model build. Three limits apply. The manipulation ran on a sibling model, not the archive’s. The padding was neutral prose, whereas the archive’s extra tokens are of unknown content and may be instruction-like. And it was added in one position, the end of the system prompt.

7. Threats to validity

Few independent units. Three LOW days in one series carry the co-variation. The day test assumes that days are exchangeable, and they are not fully: all three LOW days fall in the first week, so a trend over time could produce a similar ordering. Restricted to 21 to 27 September, when the state switched in both directions, the chance is 1 in 10.

One manipulation, on a sibling model. The archive’s state was observed and never set. The one manipulation set context size on a sibling model and found that size alone does not reproduce the pattern; it does not test what else the switch changed, and the same-model test did not run.

A seeded folder to be replaced. The seeded folder with worked entries, used in sections 6.1 and 6.2, was built partly from material that cannot be published. The paper quotes none of it, and the daily series of section 5.2 did not use the folder. Both tests are being repeated on a folder written from scratch, and those results will replace the ones in sections 6.1 and 6.2 in a later version.

A different model build behind the same name. The model name exposes no build identifier, and the output ceiling moving between 64,000 and 128,000 tokens is a serving-side change. The extra tokens may mark a different build or serving configuration rather than act as a cause. The two cannot be separated from outside, and the claim is limited to co-variation.

Unknown content. Transcripts do not keep the injected context. The identical start-up announcement, with the same tools, skills and agents, rules out the visible client configuration.

Changing client. The client moved from 2.1.278 in the archive to 2.1.291 and 2.1.292 in the later tests, and to 2.1.294 in the size manipulation.

Scope. One account, one provider, one model family and one main task. The method is portable; the finding is not shown to be.

Two instruments. The re-scorer was built after most of the daily series had run, although before the states were identified. The registered detector gives the weaker figures, and both are reported.

Boundaries set after the data. The state boundaries were chosen after the archive was seen, although no archived run on client 2.1.278 falls near them; the replication used boundaries fixed in advance.

Data not released. The run archive, the harness and the registrations are not public, so a reader cannot recompute the counts or check when each registration was committed. The checks described in appendix B were made within the project and have not been repeated by anyone outside it.

Conduct of the work. An AI agent conducted most of the experiments and analyses. The controls that do not depend on any reviewer, registration before data and recomputation from raw archives, carry more weight for that reason, and the errors found in review are reported in appendix C.

8. Recommendations for evaluations of agents

1. Record the starting context, the output ceiling and the tool list for every run. They are already present in the telemetry and cost nothing to keep.
2. Record the resolved model, not the requested name. The alias requested in this study began resolving to a different model during it.
3. Compare the fingerprints of two runs before comparing their outcomes. Runs with the same model name and the same client version were not, in this case, in the same condition.
4. Re-read conclusions by configuration state rather than by date. A date stands in for a state and is not one.
5. Treat a daily canary as dating only what it observes. The canary in this study recorded one of three switches that the runs themselves recorded.

9. Conclusion

Under one model and one client version, a coding agent’s starting configuration switched three times in sixteen days. The agent’s own telemetry recorded every switch, and a canary built for the purpose recorded one. A daily experiment registered in advance co-varied with the state, as did the way the agent read files and the format in which it wrote. Four results of the programme were measured only in one state. A replication registered in advance on the same model reproduced the large-context results, and a manipulation registered in advance showed that context size alone does not reproduce them on a sibling model. The test that would show whether the state itself matters on the archive’s model did not run while the state was observable, and is to be registered again. Independently of that test, the model name and the version string did not identify the condition under which these runs were made, and per-run telemetry, which is already recorded, came closer to identifying it.

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Appendix A. Glossary

term	meaning here
run	one headless agent session on a freshly built fixture repository, with one prompt
arm	a set of runs that share a way of delivering the rule
skill	an installed instruction file the agent may choose to open
seeded folder	a folder of past decisions placed in the fixture, with an index and two worked entries
index only	the same folder’s index, with no entries
CLAUDE.md rule	the rule written as a line in the project instruction file the client always loads
record written	the agent created a markdown file with a heading and a status value
skill opened	the harness recorded the skill’s body being read
starting context	input, cache-read and cache-creation tokens of the run’s first model call
LOW, HIGH	starting context below 26,000 tokens; at or above 29,000 tokens
canary	a fixed set of short probes run once a day from an empty directory
re-scorer	a transcript-based instrument for “record written” that reads every write route

Appendix B. How the counts were verified

The run archive, the harness, the verification scripts and the registrations are held in the author’s private repository and are not publicly released. This appendix describes how the figures in this paper were checked; it does not describe material that a reader can inspect.

Every run directory in the archive holds its provenance (date, requested and resolved model, client version, cost), detector verdicts, tool log, skill-invocation counter and full transcript. Each run’s state is assigned by a script that applies the boundaries of section 4 to the run’s first model call, and whether it wrote a record by the transcript re-scorer of section 3. Every count in this paper has a row in a claims table naming the exact runs and the kind of count, and a verification script recomputes each one from the raw archive and fails on any mismatch, or on any count in this paper without its own row. The exact intervals are checked by a separate script; the Newcombe limits, p-values and means are computed by hand. The registrations of the daily series, the alias test, the replication and the size manipulation were each committed to version control before their runs. Figure 1 is generated directly from the archive by a script.

Appendix C. Prospective tests that did not test the state

The alias test was registered in advance with five arms and requested the **sonnet** alias. Every one of its 120 runs resolved to **claude-sonnet-5-5**, a different model from the archive’s, on client 2.1.291, and every run was LOW. Its results were: bare 0/24; seeded folder with worked entries 24/24; index only 24/24; **CLAUDE.md** 24/24; skill records 20/24; skill opened 17/24, and complied 17/17 when opened. Its registered verdict was inconclusive. Because the model changed, none of these results can be attributed to the state.

The daily series’ twelfth day ran at 03:26 to 03:29 UTC on 7 October, also on **claude-sonnet-5-5**, and was LOW. Its outcome was written to the run log at 03:29. A prediction for it was committed at 14:54, in the same commit as those files, and nothing shows that the prediction was made before the outcome was seen. The day is reported as an observation on a different model, made after collection (records 6/6, skill opened 6/6), and not as a test.

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On the writing of this paper

AI tools were used in the preparation of this paper. The disclosure is offered because the paper concerns the conditions under which AI agents operate, and silence about its own production would be incongruous.

Changes in version 1.1

Version 1.1, 8 October 2026. Version 1.0, published the same day, stated that the experimental data were released with the paper, referred readers to a public archive and repository, and gave the locations of the project's files. No data or code release is being made. This version removes those statements and locations (sections 3 and 6.1, appendix B and the disclaimer) and adds the corresponding limitation to section 7.

It adds one result. Section 6 is now in two parts: 6.1 is the replication reported in version 1.0, and 6.2 is a new manipulation, registered in advance, that set the size of the starting context directly on a sibling model and found that size alone does not reproduce the large-context pattern. The abstract, the table of claims, the introduction, section 7 and the conclusion are updated to include it. The discriminating test that version 1.0 described as pending did not run while the state was observable, its launch was stopped, and it is to be registered again; the abstract and conclusion now say so. A new limitation in section 7 states that the seeded folder used in sections 6.1 and 6.2 is being replaced, and that both tests will be repeated on the new folder, with those results replacing the current ones.

It also adds robustness checks in section 5.2, states in section 3 which runs were excluded and why, corrects the introduction, which referred to negative controls that version 1.0 did not report, and corrects the account of which statistics are checked automatically. No count, interval or test result reported in version 1.0 changed.

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